

KENDRIYA VIDYALAYA SANGATHAN (PATNA REGION)

PRE-BOARD 1 EXAMINATION (2024-25)

CLASS X (SET B)

(Science 086)

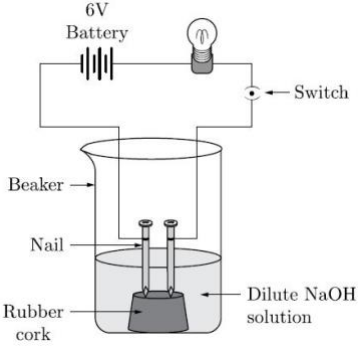
Max. Marks: 80

Time Allowed: 3 hours

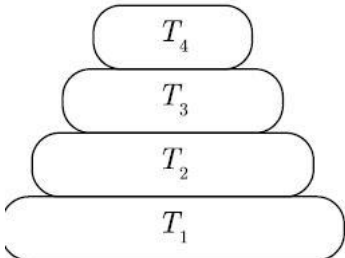
General Instructions:

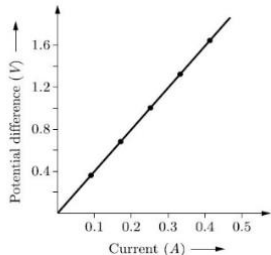
- 1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.**
- 2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.**
- 3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.**
- 4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.**
- 5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.**
- 6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.**

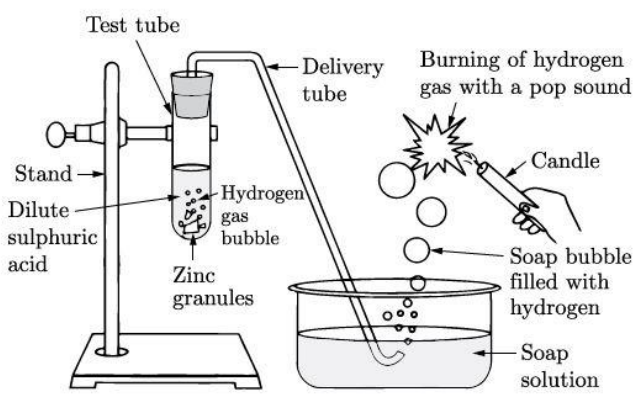
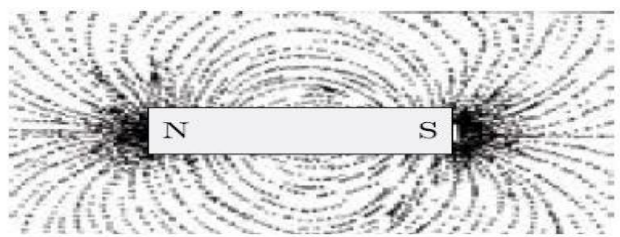
	Section-A	
	Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.	
1	Select from the following a decomposition reaction in which source of energy for decomposition is light: (a) $2\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$ (b) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ (c) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$ (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$	1
2	An electrolytic cell consists of (i) positively charged cathode. (ii) negatively charged anode (iii) positively charged anode. (iv) negatively charged cathode (a) (i) and (ii) (b) (iii) and (iv) (c) (i) and (iii) (d) (i) and (ii)	1

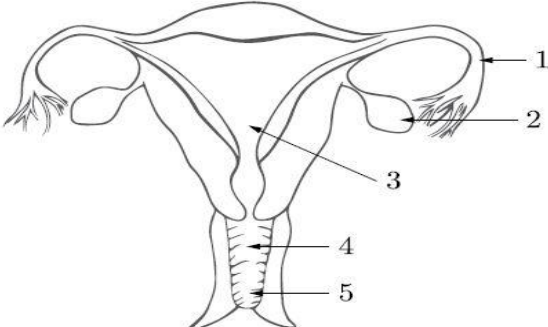
3	Reaction between X and Y, forms compound Z. X loses electron and Y gains electron. Which of the following properties is not shown by Z? (a) Has high melting point. (b) Has low melting point (c) Conducts electricity in molten state. (d) Occurs as solid	1
4	In an attempt to demonstrate electrical conductivity through an electrolyte, the following apparatus (Figure) was set up.  Which among the following statement(s) is(are) correct ? (i) Bulb will not glow because electrolyte is not acidic (ii) Bulb will glow because NaOH is a strong base and furnishes ions for conduction. (iii) Bulb will not glow because circuit is incomplete (iv) Bulb will not glow because it depends upon the type of electrolytic solution (a) (i) and (iii) (b) (ii) and (iv) (c) (ii) only (d) (iv) only	1
5	In the double displacement reaction between aqueous potassium iodide and aqueous lead nitrate, a yellow precipitate of lead iodide is formed. While performing the activity if lead nitrate is not available, which of the following can be used in place of lead nitrate? (a) Lead sulphate (insoluble) (b) Lead acetate (c) Ammonium nitrate (d) Potassium sulphate	1
6	The electronic configurations of three elements X, Y and Z are X - 2, 8 ; Y - 2, 8, 7 and Z - 2, 8, 2. Which of the following is correct? (a) X is a metal (b) Y is a metal (c) Z is a non-metal (d) Y is a non-metal and Z is a metal	1
7	Which among the following statement(s) is(are) true? Exposure of silver chloride to sunlight for a long duration turns grey due to: (i) the formation of silver by decomposition of silver chloride (ii) sublimation of silver chloride (iii) decomposition of chlorine gas from silver chloride (iv) oxidation of silver chloride	1

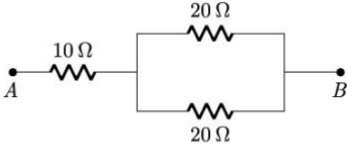
	(a) (i) only. (b) (i) and (iii) (c) (ii) and (iii). (d) (iv) only	
8	Choose the event that does not occur in photosynthesis (a) Absorption of light energy by chlorophyll (b) Reduction of carbon dioxide to carbohydrates (c) Oxidation of carbon to carbon dioxide (d) Conversion of light energy to chemical energy	1
9	Choose the correct statement that describes arteries. (a) They have thick elastic walls, blood flows under high pressure; collect blood from different organs and bring to the heart. (b) They have thin walls with valves inside, blood flows under low pressure and carry blood away from the heart to various organs of the body. (c) They have thick elastic walls, blood flows under low pressure; carry blood from the heart to various organs of the body. (d) They have thick elastic walls without valves inside, blood flows under high pressure and carry blood away from the heart to different parts of the body.	1
10	The breakdown of pyruvate to give carbon dioxide, water and energy takes place in (a) cytoplasm. (c) chloroplast. (b) mitochondria. (d) nucleus.	1
11	The growth of tendril in pea plants is due to (a) effect of light (b) effect of gravity (c) rapid cell divisions in tendrillar cells that are away from the support (d) rapid cell divisions in tendrillar cells in contact with the support	1
12	Select the statements that describe characteristics of genes (i) genes are specific sequence of bases in a DNA molecule (ii) a gene does not code for proteins (iii) in individuals of a given species, a specific gene is located on a particular chromosome (iv) each chromosome has only one gene (a) (i) and (ii) (b) (i) and (iii) (c) (i) and (iv) (d) (ii) and (iv)	1
13	In which of the following, the image of an object placed at infinity will be highly diminished and point sized? (a) Concave mirror only (b) Convex mirror only (c) Convex lens only (d) Concave mirror, convex mirror, concave lens and convex lens	1
14	Rays from Sun converge at a point 15 cm in front of a concave mirror. Where should an object be placed so that size of its image is equal to the size of the object ? (a) 15 cm in front of the mirror (b) 30 cm in front of the mirror	1

	(c) between 15 cm and 30 cm in front of the mirror (d) more than 30 cm in front of the mirror	
15	In the given figure the various trophic levels are shown in a pyramid. At which trophic level is maximum energy available?  (a) T4 (b) T2 (c) T1 (d) T3	1
16	The decomposer's in an ecosystem (a) convert inorganic material, to simpler forms (b) convert organic material to inorganic forms (c) convert inorganic materials into organic compounds (d) do not breakdown organic compounds	1
	Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	
17	Assertion: When zinc is added to a solution of iron (II) sulphate, no change is observed. Reason: Zinc is more reactive than iron.	1
18	Assertion: Dominant allele is an allele whose phenotype expresses even in the presence of another allele of that gene. Reason: It is represented by a capital letter, e.g. T.	1
19	Assertion: A ray of light incident along the normal to the plane mirror retraces its path after reflection from the mirror. Reason: A ray of light along the normal has angle of incidence as 90° and hence, it retraces its own path after reflection from mirror.	1
20	Assertion: Herbivores are called first order consumers. Reason: Tiger is a top carnivore.	1
	SECTION-B Question No. 21 to 26 are very short answer questions	
21	Translate the following statements into chemical equations and then balance them : (i) Carbon disulphide burns in air to give carbon dioxide and sulphur dioxide. (ii) Sodium metal reacts with water to give sodium hydroxide and hydrogen gas. (iii) Aluminium metal replaces hydrogen from sulphuric acid and gives aluminium sulphate. (iv) Ferric oxide oxidises CO to CO_2 and itself gets reduced to Fe.	2

22	A certain tissue in a green plant somehow got blocked and the roots were deprived of food. What was the tissue that got affected?	2
23	Name the process by which an amoeba reproduces. Draw the various stages of its reproduction in a proper sequence. OR A student is viewing under a microscope a permanent slide showing various stages of asexual reproduction by budding in yeast. Draw diagram of what he observes in proper sequence.	2
24	An object of height 4.0 cm is placed at a distance of 30 cm from optical centre 'O' of a convex lens of focal length 20 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre 'O' and principal focus 'F' on the diagram. Also find the approximate ratio of size of image to the size of object.	2
25	V - I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.  OR Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it? State the law which helps us in solving the above stated question.	2
26	Which one of the following food habits is better and why ? (a) Plant----→ Man. (b) Plant----→ Goat----→ Man	2
SECTION –C		
Question No. 27 to 33 are short answer questions		
27	A metal M reacts vigorously with water to form a solution S and a gas G. The solution S turns red litmus to blue whereas gas G, which is lighter than air, burns with a pop sound. Metal M has a low melting point and is used as a coolant in nuclear reactors. (i) What is metal M? (ii) What is solution S? (iii) What is gas G?	3
28	In the schematic diagram given below for the preparation of hydrogen gas what would happen if following changes are made? (a) In place of zinc granules, same amount of zinc dust is taken in the test tube. (b) Instead of dilute sulphuric acid, dilute hydrochloric acid is taken.	3

	(c) Sodium hydroxide is taken in place of dilute sulphuric acid and the tube is heated.  OR Why the medium becomes acidic in mouth? What is the ill effect of the acidic medium? How this can be prevented?	
29	What are the events taking place during photosynthesis?	3
30	Name the plant Mendel used for his experiment. What type of progeny was obtained by Mendel in F1 and F2 generations when he crossed the tall and short plants? Write the ratio he obtained in F2 generation plants.	3
31	What is "dispersion of white light"? Draw a labelled diagram to illustrate the recombination of the spectrum of white light. Why is it essential that the two prisms used for the purpose should be identical and placed in an inverted position with respect to each other?	3
32	A piece of wire of resistance 20 ohm is drawn out so that its length is increased to twice its original length. Calculate the resistance of the wire in the new situation.	3
33	Study the diagram given below and answer the following questions:  (a) Why do the iron filings arrange in such a pattern? (b) What does this pattern demonstrate? (c) Why do the iron filings near the bar magnet seem to align in the shape of closed curves?	3
	SECTION-D	
	Question No. 34 to 36 are long answer questions.	
34	The formulae of four organic compounds A, B, C and D are given below: A- C₂H₄, B- CH₃COOH, C- C₂H₅OH, D- C₂H₆	5

	(i) Which one of these compounds A, B, C or D is a saturated hydrocarbon? (ii) Identify the organic acid and give its structural formula. (iii) Which of the above compounds when heated at 443 K in the presence of concentrated H_2SO_4 forms ethene as the major product? What is the role played by concentrated H_2SO_4 in this reaction? Also write the chemical equation involved. (iv) Give a chemical equation when B and C react with each other in presence of concentrated H_2SO_4. Name the major product formed and mention one of its important use. OR (a) Give a chemical test to distinguish between saturated and unsaturated hydrocarbons. (b) Name the products formed when ethane burns in air. Write the balanced chemical equation for the reaction showing the types of energies liberated. (c) Why is reaction between methane and chlorine in the presence of sunlight considered a substitution reaction?	
35	(a) Write the functions of the following parts in human female reproductive system: (i) Ovary, (ii) Oviduct, (iii) Uterus (b) Describe the structure and function of placenta. OR (a) Identify the given diagram. Name the parts 1 to 5.  (b) What is contraception? List three advantages of adopting contraceptive measures.	5
36	(a) Three resistors R_1, R_2 and R_3 are connected in parallel and the combination is connected to a battery, ammeter, voltmeter and key. Draw suitable circuit diagram and obtain an expression for the equivalent resistance of the combination of the resistors. (b) Calculate the equivalent resistance of the following network:	5

	 OR Show how would you join three resistors, each of resistance 9 Ohm so that the equivalent resistance of the combination is: (a) 13.5 Ohm (b) 6 Ohm	
	SECTION-E Question No. 37 to 39 are case-based/data -based questions.	
37	Read the following case based passage and answer the questions given after passage. Structural formulae of some organic compounds are given below and every organic compound have different structural formulae. (A) $\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C}- & \text{C}-\text{H} \\ & \\ \text{H} & \text{H} \end{array}$ (C) $\begin{array}{c} \text{H} & & \text{H} \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{H} & & \text{H} \end{array}$ (E) $\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{H} \end{array}$ (B) $\text{H}-\text{C}\equiv\text{C}-\text{H}$ (D) $\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$ (F) $\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C}- & \text{C}-\text{OH} \\ & \\ \text{H} & \text{H} \end{array}$ (i) Name the compounds which can undergo addition reactions. (ii) How can F be converted to C? (iii) What is the name of functional group of compound F? (iv) Which compounds belong to same homologous series in given organic compound?	4
38	Nastic movements in plants are not directional movements. They are not dependent on the stimulus and are growth independent. For example, the leaves of a touch me not plant (<i>Mimosa pudica</i>), fold up immediately when touched. These kinds of changes occur due to the changes in the amount of water in the leaves. Depending on the quantity, they either swell up or shrink. Plant hormones or phytohormones are responsible for the control and coordination of plants. There are different types of hormones, which affect the growth of a plant. Phytohormones are chemical compounds which are released by stimulated cells. These hormones are diffused around the plant cells. They have a role in the cell division, cell enlargement, cell differentiation, fruit growth, falling of leaves, ripening of fruits, ageing of plants etc.	4

	 (i) Name the phenomenon called for the movement in growth of plants. (ii) What do you mean by nastic movement? (iii) What are the different types of harmonies of plants? (iv) The plant hormone help in the cell growth at the shoot tips by elongating the cells and help in the growth process is : _____	
39	Convex mirror is used as a rear-view mirror in vehicles. Since the image of the object formed is small in size, the field of view is increased. Convex mirror is also used in streetlights to diverge light over a large area.  (i) In driver's mirror, what type of image is formed behind the vehicle? (ii) What can you say about field of view of a convex mirror? (iii) A convex mirror is used to form the image of an object. What is the nature of formed image? (iv) How can we identify a convex mirror by touch? OR (i) Which type of mirror is used in headlight of the car? (ii) Why the convex mirror is called diverging mirror? (iii) Which mirror is used in solar cooker and why? (iv) Where the object to be kept to obtain an enlarged and virtual image in a concave mirror?	4